ENVS 193DS Homework 3

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Homework Setup

```
# reading in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
# creating new object from data set
kelp <- read_csv(here("data","temp-kelp.csv"))
bike <- read_csv(here("data","ENVS-193DS-Personal-Data-Project-cont.csv"))
```

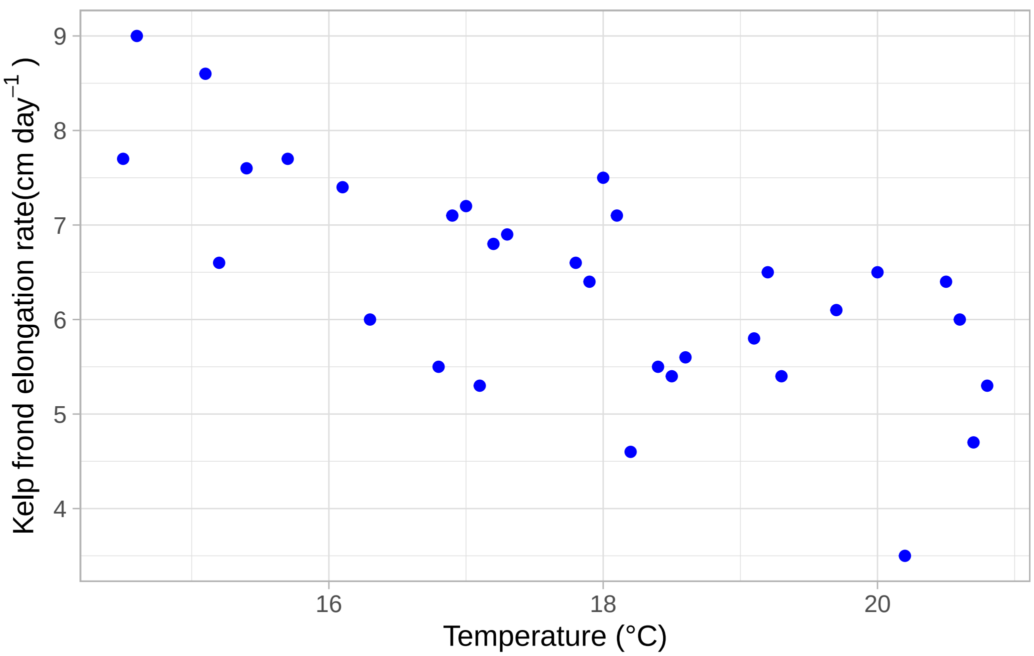
Problem 1. Giant kelp fronds

a. An appropriate test

The two tests that are appropriate to describe the strength of the relationship between temperature and giant kelp frond elongation rate are a linear model and a correlation test. A linear model will tell us how well temperature can be used to predict kelp frond elongation rate, while correlation can tell us whether these two variables are related to each other. Given that we do not have a specific hypothesis on how temperature affects kelp frond elongation rate, both tests should be used to accurately describe the relationship.

b. Create a visualization

```
# base layer: ggplot
ggplot(data = kelp, # data frame: kelp
       # x-axis: temperature
       # y-axis: kelp frond elongation rate
       mapping = aes(x = temp_c,
                     y = kelp_elong)) +
  # first layer: points to represent individual observations
  geom_point(color = "blue") + # using custom colors
  # relabeling axes, adding title
  labs(x = "Temperature (°C)",
       y = expression("Kelp frond elongation rate(cm day"^-1~")")) +
  # changing theme
  theme_light()
```

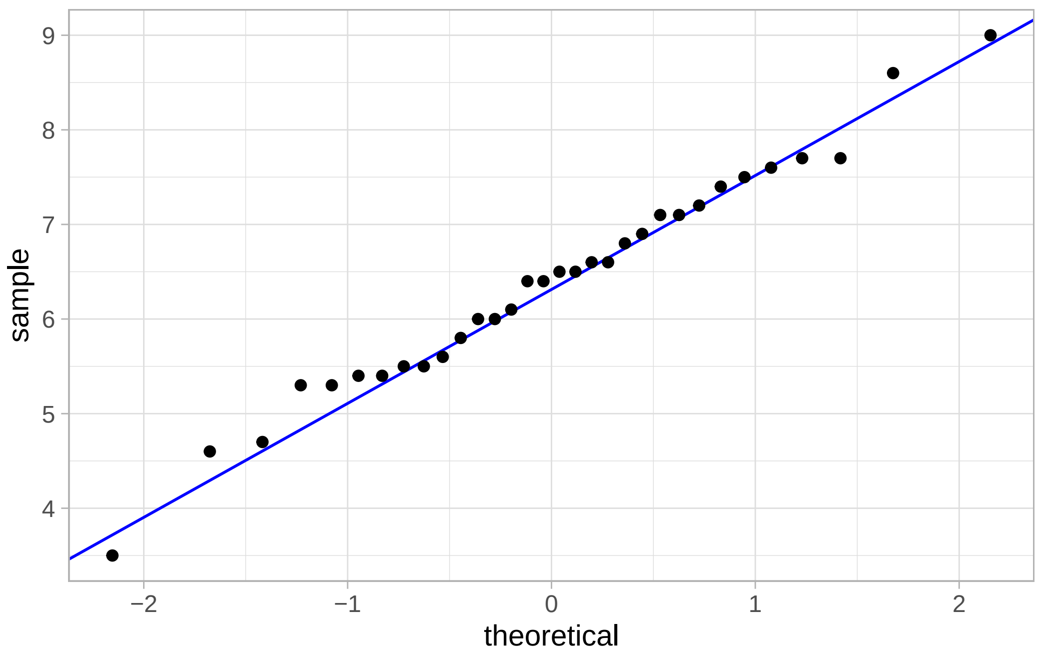


c. Check your assumptions and run your test

Checking assumptions

```
# base layer: ggplot
ggplot(data = kelp, # data frame: kelp
```

```
      # y-axis for QQ plot (no x-axis for QQ plot)
      aes(sample = kelp_elong)) +
  # first layer: QQ reference line
  geom_qq_line(color = "blue") +
  # second layer: QQ plot
  geom_qq() +
  # custom theme
  theme_light()
```



Based on the initial visualization, we can expect a linear relationship between temperature and kelp frond elongation rate. Both temperature and kelp frond elongation rate are continuous variables. Both variables are normally distributed, based on points roughly following the reference line on the QQ plot. Finally, observations are assumed to be independent.

Running test

```
cor.test(kelp$kelp_elong, kelp$temp_c, # variables: kelp elongation rate and temperature
         method = "pearson") # method: pearson's correlation (r)
```

```
    Pearson's product-moment correlation

data:  kelp$kelp_elong and kelp$temp_c
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

d. Results communication

To evaluate the strength of the relationship between temperature and giant kelp frond elongation rate, we used a Pearson's correlation (r) test. A correlation test was appropriate for this as we wanted to determine the strength of the relationship between these two variables.

We found a moderate negative relationship between ocean temperature and giant kelp frond elongation rate (Pearson's r = -0.69, t(30) = -5.12, $p < 0.001$, $\alpha = 0.05$).

e. Test implications

From this study, we found that as ocean temperature increases, giant kelp frond elongation rate decreases. We found a moderate negative relationship between ocean temperature and giant kelp frond elongation rate (Pearson's r = -0.69, t(30) = -5.12, $p < 0.001$, $\alpha = 0.05$).

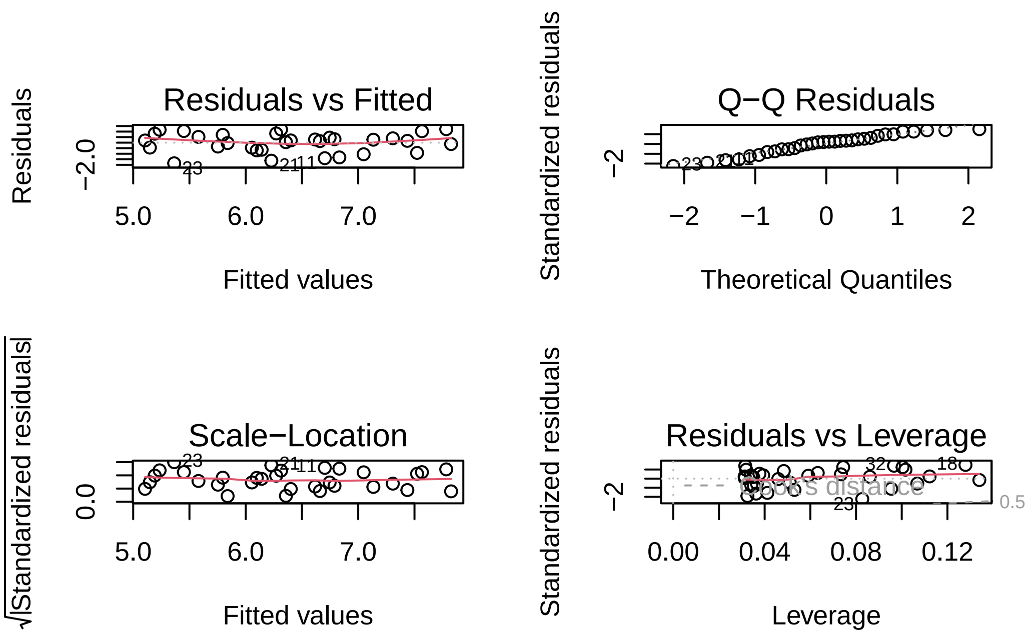
f. Double check your work

Fitting linear model

```r
kelp_model <- lm(
  kelp_elong ~ temp_c, # formula: kelp elongation rate as a function of temperature
  data = kelp # data frame: kelp
)
```

Diagnostics

```r
# display plots in a 2x2 grid
par(mfrow = c(2, 2))
# display diagnostic plots
plot(kelp_model)
```



- Residuals vs Fitted and Scale-Location: checking homoscedasticity of residuals, residuals seem to have constant variance based on even distribution around the horizontal red line
- Q-Q Residuals: checking distribution of residuals, residual is normally distributed based on points following the dotted line relatively closely
- Residuals vs Leverage: checking for outliers, no outliers outside dotted lines that would influence model predictions

Model Summary

```r
# displaying result of kelp linear model
summary(kelp_model)
```

```
Call:
```

```
lm(formula = kelp_elong ~ temp_c, data = kelp)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.8643 -0.5017  0.1790  0.5659  1.2177 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 14.08645    1.49219   9.440 1.72e-10 ***
temp_c      -0.43179    0.08321  -5.189 1.37e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom
Multiple R-squared:  0.473,	Adjusted R-squared:  0.4554 
F-statistic: 26.93 on 1 and 30 DF,  p-value: 1.366e-05
```

For each 1°C of ocean temperature increase, we would expect a giant kelp frond elongation rate decrease of 0.43 $\pm$ 0.08 (SE) cm/day.

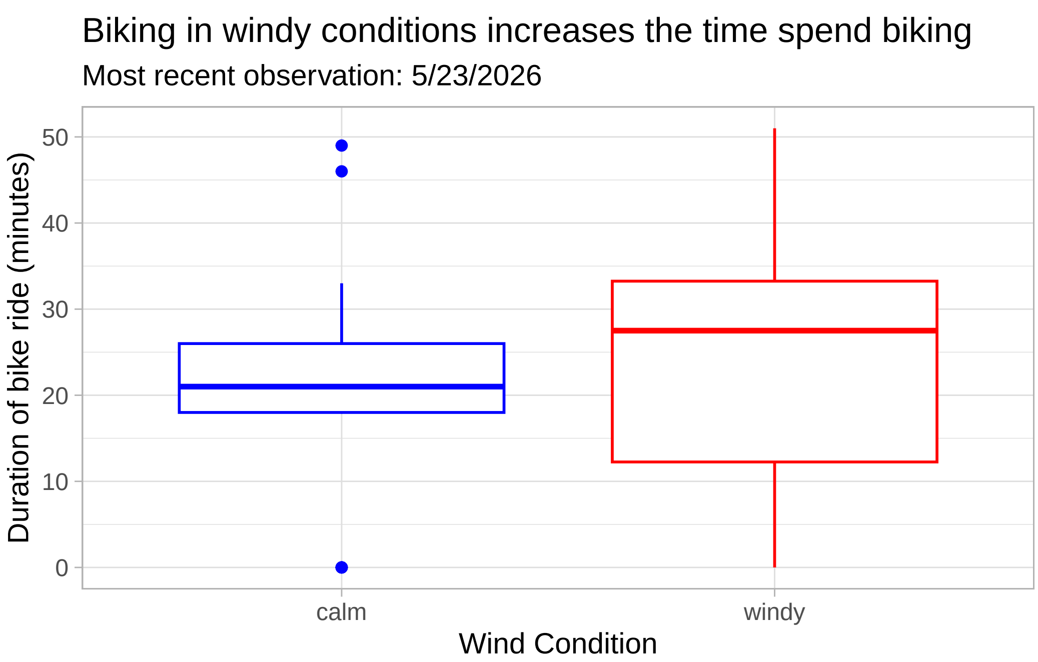
This result from the linear model would have led me to make the same decision about the null hypothesis. As ocean temperature increases, giant kelp frond elongation rate decreases. The only difference is that the strength of this relationship would have not been known without the correlation test.

Problem 2. Personal Data

a. Updating your visualizations

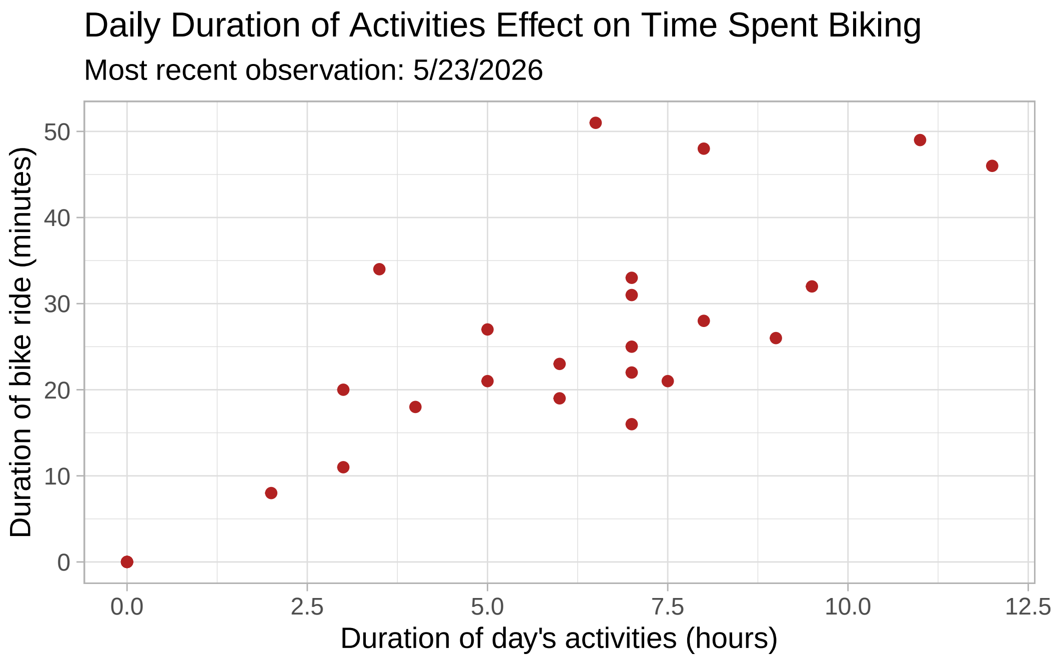
```
# base layer: ggplot
ggplot(data = bike, # data frame: bike
       # x-axis: weather
       # y-axis: Duration of bike ride
       # coloring by weather
       aes(x = Weather,
           y = `Duration of bike ride (minutes)`,
       color = Weather)) +
  # first layer: boxplot
  geom_boxplot() +
 # relabeling the axes, adding title and subtitle
```

```
  labs(x = "Wind Condition",
       y = "Duration of bike ride (minutes)",
       title = "Biking in windy conditions increases the time spend biking",
       subtitle = "Most recent observation: 5/23/2026") +
 # custom theme
  theme_light() +
  # removing legend
  theme(legend.position = "none") +
  # changing the color
  scale_color_manual(values = c("calm" = "blue",
                                "windy" = "red"))
```




```
# base layer: ggplot
ggplot(data = bike, # data frame: bike
       # x-axis: duration of day's activities
       # y-axis: Duration of bike ride
       aes(x = `Duration of day's activities (hours)`,
           y = `Duration of bike ride (minutes)`,)) +
  # first layer: points
  geom_point(color = "firebrick") +
 # relabeling the axes, adding title and subtitle
  labs(x = "Duration of day's activities (hours)",
```

```
      y = "Duration of bike ride (minutes)",
      title = "Daily Duration of Activities Effect on Time Spent Biking",
      subtitle = "Most recent observation: 5/23/2026") +
# custom theme
 theme_light() +
 # removing legend
 theme(legend.position = "none")
```



b. Captions

Figure 1. Biking in windy conditions increases the time spend biking Difference in time spent biking between calm and windy conditions is shown as a boxplot. Median, IQR, errorbars, and outliers are visualized and wind condition type is differentiated based on color. Wind data citation: Weather Underground SBFD Station 1 - KCASANTA4905

Figure 2. Daily Duration of Activities Effect on Time Spent Biking As duration of day's activities increases, time spent biking also increases. Individual, daily observations are shown as dots on the graph.

Problem 3. Affective visualization

a. Describe in words what an affective visualization could look like for your personal data.

b. Create a sketch of your idea

c. Make a draft of your visualization

d. Write an artist statement

e. Prep your materials to share in class

Problem 4. Statistical critique

a. Revisit and summarize

The authors tested the hypothesis that California abalone species are negatively affected by warming ocean temperatures. - Test used: ANOVA (Analysis of Variance) - Response Variable(s): Abalone growth, reproductive output, disease, mortality - Predictor Variable(s): Temperature, food quantity, food quality The authors would interpret a significant result ($p < 0.05$) as there is a difference in average abalone growth, reproduction, disease, and mortality between the two study species, *H. rufescens* and *H. fulgens*.

TABLE 1. F values and probabilities from three-factor ANOVAs with equal replication, testing the effects of varying temperature, food quantity, and food quality on abalone growth, reproductive output, disease, and mortality.

		Growth								
		Length			Mass			Reproduction (GBI)		
Source of variation	df	MS	F	P	MS	F	P	MS	F	P
Haliotis rufescens										
Temperature	2	0.05	181.87	0.00	1.80	276.15	0.00	0.22	3.87	0.03
Quantity	2	0.08	272.42	0.00	1.40	215.86	0.00	1.86	32.51	0.00
Quality	2	0.02	79.22	0.00	0.51	79.08	0.00	0.50	8.74	0.00
Temperature : Quantity	4	0.00	4.16	0.01	0.02	3.71	0.01	0.05	0.90	0.47
Temperature : Quality	4	0.00	4.76	0.00	0.03	5.30	0.00	0.03	0.55	0.70
Quantity : Quality	4	0.00	10.22	0.00	0.06	9.55	0.00	0.29	5.14	0.00
Temperature : Quantity : Quality	8	0.00	0.34	0.94	0.01	1.06	0.41	0.03	0.59	0.78
Residuals	54	0.00			0.01			0.06		
H. fulgens										
Temperature	2	0.00	11.34	0.00	0.01	2.41	0.10	0.84	8.05	0.00
Quantity	2	0.12	690.91	0.00	1.57	474.76	0.00	7.32	70.11	0.00
Quality	2	0.02	133.68	0.00	0.29	88.96	0.00	0.36	3.45	0.04
Temperature : Quantity	4	0.00	9.10	0.00	0.05	14.36	0.00	0.52	4.99	0.00
Temperature : Quality	4	0.00	1.46	0.23	0.00	0.85	0.50	0.13	1.24	0.30
Quantity : Quality	4	0.00	22.46	0.00	0.03	10.10	0.00	0.05	0.47	0.76
Temperature : Quantity : Quality	8	0.00	0.82	0.59	0.00	1.08	0.39	0.08	0.78	0.62
Residuals	54	0.00			0.00			0.10		

Notes: Growth was evaluated through standardized changes in animal mass and shell length [(initial − final value)/initial value]. Reproductive output was evaluated from gonad bulk indices (GBI). Disease was evaluated by the burden of withering syndrome, Rickettsiales-like prokaryote (WS-RLP) and scores of morphological changes associated with withering syndrome (digestive gland atrophy [DGA] and foot muscle degeneration [FM-DG]), and accumulated mortalities. All variables were derived from cell means since that was the experimental unit and not individual abalone (see *Methods* for details). A posteriori tests were run on those treatments with significant effects (see Appendix).

TABLE 1. Extended.

Disease and morbidity											
WS-RLP			DGA			FM-Deg			Mortality		
MS	F	P	MS	F	P	MS	F	P	MS	F	P
4.74	19.64	0.00	2.64	25.32	0.00	1.80	9.48	0.00	0.38	41.11	0.00
0.34	1.40	0.25	1.78	17.07	0.00	2.08	10.90	0.00	0.04	3.97	0.02
0.07	0.28	0.76	0.40	3.84	0.03	0.98	5.13	0.01	0.00	0.08	0.92
0.19	0.79	0.54	0.32	3.05	0.02	0.15	0.81	0.53	0.03	3.49	0.01
0.10	0.41	0.80	0.15	1.43	0.24	0.05	0.25	0.91	0.00	0.20	0.94
0.31	1.29	0.28	0.02	0.23	0.92	0.24	1.24	0.30	0.00	0.16	0.96
0.35	1.45	0.20	0.12	1.14	0.35	0.08	0.43	0.90	0.00	0.34	0.94
0.24			0.10			0.19			0.01		
0.02	0.86	0.43	0.00	1.00	0.37	0.08	1.19	0.31	0.00	2.00	0.15
0.00	0.21	0.81	0.00	1.00	0.37	0.05	0.81	0.45	0.00	0.50	0.61
0.02	0.86	0.43	0.00	1.00	0.37	0.26	3.94	0.03	0.00	0.50	0.61
0.02	1.07	0.38	0.00	1.00	0.42	0.03	0.50	0.74	0.00	0.50	0.74
0.03	1.71	0.16	0.00	1.00	0.42	0.07	1.09	0.37	0.00	0.50	0.74
0.02	1.07	0.38	0.00	1.00	0.42	0.04	0.63	0.65	0.00	1.25	0.30
0.01	0.64	0.74	0.00	1.00	0.45	0.05	0.69	0.70	0.00	1.25	0.29
0.02			0.00			0.07			0.00		

b. Visual clarity

c. Aesthetic clarity

d. Recommendations